

Winston Yap

Ezra Systems Postdoctoral Associate, Cornell University | PhD' 2025 National University of Singapore
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RESEARCH AREAS

- Urban Complexity and Systems Thinking
- Urban Health and Sustainability
- Network Science and Graph Machine Learning
- Open Source and Globally Scalable Urban Analytics

EDUCATION

National University of Singapore, Singapore

Ph.D. in Urban Analytics

Nov 2025

- Cumulative GPA: 4.75/5.00
- Advisor: Prof. Filip Biljecki
- Thesis title: Urban Graph Analytics - Connecting Cities, Data, and People

National University of Singapore, Singapore

Master of Urban Planning

Apr 2018

- 95th Percentile

National University of Singapore, Singapore

Bachelor of Science in Real Estate (Hons.)

Apr 2017

- Specialisation in Urban Planning

RESEARCH EXPERIENCE

Ezra Systems Postdoctoral Associate

Feb 2026 – Now

Systems Engineering, Cornell University, Ithaca, NY, USA

Advisors: Prof. Oliver H. Gao

- Human-AI Urban Systems Intelligence

Visiting Scholar

Jul 2024 – Feb 2025

Senseable City Laboratory, Massachusetts Institute of Technology, Cambridge, MA, USA

Advisors: Prof. Carlo Ratti and Dr. Fábio Duarte

- Quantifying Health Inequities with Large-Scale Urban Data and GeoAI
- Assessing Spatio-Temporal Network Growth in Chinese Cities
- Generative Geospatial Visualization and Narrative Generation with LLMs

PhD Industry Intern, ML Engineer

Jan 2024 – Jul 2024

Map IV, Inc., Spatial Technology Team, Nagoya, Japan

- High Definition Mapping with 3D Deep Learning

Graduate Research Associate

Apr 2018 – Aug 2021

LKYCIC, Singapore University of Technology and Design, Singapore

Advisors: Prof. Chan Heng Chee, Prof. Harvey Neo, Prof. Belinda Yuen

- Planning and Design of Age-Friendly Neighbourhoods in Singapore
- Comparative Analysis of Urbanization in Southeast Asian Megacities
- Investigating Socio-spatial Reconfiguration of Urban Cores Post-COVID

Graduate Research Assistant

Apr 2017 – Aug 2017

Deguchi Lab, University of Tokyo, Kashiwanoha, Japan

- Urban Revitalization Planning with Local Communities

Summer Research Intern

Apr 2016 – Aug 2016

Housing and Development Board, Ministry of National Development, Singapore

- Study of Industrial Planning Strategies and Design Criteria

ACADEMIC HONORS & AWARDS

SMU Overseas Postdoctoral Fellowship College of Integrative Studies, Singapore Management University	2026
Department Travel Award for Attending CUPUM 2025 at London Department of Architecture, National University of Singapore	2025
WCS Young Leader (Nominated from diverse partners and networks) World Cities Summit, Ministry of National Development, Singapore	2024
IFoU Young Ambassador (Nominated by executive committee) International Forum on Urbanism	2024
NUS Overseas Research Immersion Award (Merit-based, fund MIT exchange) College of Design and Engineering, National University of Singapore	2024
NUS Overseas College Programme (1 of 2 PhD students selected from NUS) NUS Enterprise, National University of Singapore	2023
PhD Dissertation Research Fellowship (National Top 10 Data Science PhD) Singapore Data Science Consortium, National Research Foundation	2023
Asia Travel Award for Attending FOSS4G Asia Conference at Seoul, South Korea Open Source Geospatial Foundation	2023
NUS PhD Research Scholarship (Merit-based, support 4-year PhD study) College of Design and Engineering, National University of Singapore	2021
Asian Masters of Urbanism Certificate (1 of 2 Master's students from cohort) Department of Architecture, National University of Singapore	2017

PUBLICATIONS

UNDER REVIEW / IN PREPARATION

[23] **Yap, W.**, Sy, C., Yang, H., Liu, X., Smith, K., Duarte, F., Zheng, Y., Ratti, C., Biljecki, F., Gao, O. Dynamic Human-AI Collaboration for Decision-Making in Complex Urban Systems, (2026). In preparation.

[22] Liu, P., **Yap, W.**, Hou, Y., Lei, B., Stouffs, R., Biljecki, F. A Graph-Based Human-Centred GeoAI for Understanding Street-Level Environment and Traffic Accident Frequency with Street View Imagery, (2025). Accepted, *Journal of Urban Technology*.

[21] Torkko, J., Quintana, M., Willberg, E., Malekzadeh, M., Yang, S., **Yap, W.**, Korpilo, S., Gu, Y., Biljecki, F., Toivonen, T. (2026). Submitted to *Landscape and Urban Planning*.

[20] Fan, Z., **Yap, W.**, Law, S., Chen, Y., Li, J., Biljecki, F. (2026). Submitted to *By Looks or by Form: Disentangling Streetscape Appearance and Geometry to Read Urban Wealth and Activity Gaps*. Preprint submitted.

[19] Zhou, SS., Tao, L., Wei, H., **Yap, W.**, ..., Yang H. Emergent Capacity of Urban Low-Altitude Airspace Under Evolutionary Operator Strategies. (2026). Submitted to *npj Urban Sustainability*.

[18] Luo, P., **Yap, W.**, Ratti, C., Santi, P., Duarte, F. Spatio-Temporal Topological Change in Chinese Cities, (2025). In preparation.

[17] **Yap, W.**, Strobel, L., Pruckner, M. Mapping Urban Structure and Inequality Through Semantic Mobility Patterns

in Multiple Cities, (2025). In preparation.

PEER-REVIEWED JOURNAL PAPERS

- [16] **Yap, W.**, Duarte, F., Zheng, Y., Jang, K.M., Luo, P., Vineis, P., Ratti, C., Biljecki, F. Urban motifs associated with population health. *Nature Health*. (2026). <https://doi.org/10.1038/s44360-026-00151-9>
- [15] Liang, X., **Yap, W.**, Biljecki, F. Heterogeneous graph neural networks for building attribute prediction from hierarchical urban features and cross-view imagery. 234, 185-204. *ISPRS Journal of Photogrammetry and Remote Sensing*. (2026).
- [14] Ma, R., Cao, B., **Yap, W.**, Cao, N., Ratti, C., Duarte, F. GeoNarrative - Transforming Geospatial Textual Data into Narrative Storytelling Visualization. *IEEE VIS 2026*.
- [13] Fujiwara, K., Tsurumi, R., Kiyono, T., Fan, Z., ... **Yap, W.**, Biljecki, F. VoxCity: A Seamless Framework for Open Geospatial Data Integration, Grid-Based Semantic 3D City Model Generation, and Urban Environment Simulation. *Computers, Environment, and Urban Systems*, 123, 102366, (2026).
- [12] Moser, M., Resch, B., Zeile, P., Keskin, M., Schmidt, S., **Yap, W.**, Graf, D., Heinke, M., Gandhi, S. Understanding the Influence of Urban Characteristics on Cyclists' Stress Measured through Wearable Sensors: A Quantitative Open Data Approach. *Environment and Planning B: Urban Analytics and City Science*. 0-0, (2025).
- [11] **Yap, W.**, Wu, A. N., Miller, C., Biljecki, F. Revealing building operating carbon dynamics for multiple cities. *Nature Sustainability*, 1-12, (2025). <https://doi.org/10.1038/s41893-025-01615-8>.
- [10] Gu, Y., Quintana, M., Liang, X., Ito, K., **Yap, W.**, Biljecki, F. Designing effective image-based surveys for urban visual perception. *Landscape and Urban Planning*, 260, 105368, (2025).
- [9] Fujiwara, K., Khomiakov, M., **Yap, W.**, Ignatius, M., Biljecki, F. Microclimate Vision: Multimodal prediction of climatic parameters using street-level and satellite imagery. *Sustainable Cities and Society*, 114, 105733, (2024).
- [8] Hou, Y., Quintana, M., Khomiakov, M., **Yap, W.**, Ouyang, J., Ito, K., ... Biljecki, F. Global Streetscapes—A comprehensive dataset of 10 million street-level images across 688 cities for urban science and analytics. *ISPRS Journal of Photogrammetry and Remote Sensing*, 215, 216-238, (2024).
- [7] Wang, S., Huang, X., Liu, P., Zhang, M., ..., **Yap, W.**, Bao, S. Mapping the landscape and roadmap of geospatial artificial intelligence (GeoAI) in quantitative human geography: An extensive systematic review. *International Journal of Applied Earth Observation and Geoinformation*, 128, 103734, (2024).
- [6] **Yap, W.** Biljecki, F. A Global Feature-Rich Network Dataset of Cities and Dashboard for Comprehensive Urban Analyses. *Scientific Data*, 10, 667, (2023). <https://doi.org/10.1038/s41597-023-02578>
- [5] **Yap, W.**, Stouffs, R. Biljecki, F. Urbanity: automated modelling and analysis of multidimensional networks in cities. *npj Urban Sustainability*, 3, 45, (2023). <https://doi.org/10.1038/s42949-023-00125-w>
- [4] **Yap, W.**, Chang, J. H., Biljecki, F. Incorporating networks in semantic understanding of streetscapes: Contextualising active mobility decisions. *Environment and Planning B: Urban Analytics and City Science*, 50, 1416-1437, (2022). <https://doi.org/10.1177/2399808322113883>
- [3] **Yap, W.**, Janssen, P., Biljecki, F. Free and open source urbanism: software for urban planning practice. *Computers, Environment and Urban Systems*, 96, 101825, (2022). <https://doi.org/10.1016/j.compenvurbsys.2022.10182>
- [2] Hou, Y., **Yap, W.**, Chua, R., Song, S., Yuen, B. The associations between older adults' daily travel pattern and objective and perceived built environment: A study of three neighbourhoods in Singapore. *Transport Policy*, 99, 314-328, (2020).
- [1] Song, S., **Yap, W.**, Hou, Y., Yuen, B. Neighbourhood built Environment, physical activity, and physical health among older adults in Singapore: A simultaneous equations approach. *Journal of Transport & Health*, 18, 100881, (2020).

CONFERENCE AND WORKSHOP PAPERS

- [5] **Yap, W.** Gao, O. Eliciting Causal Urban System Intelligence through AI-Powered Knowledge Graphs. *Conference on Complex Systems*, (2026, October).
- [4] **Yap, W.** Biljecki, F. Leveraging Urban Graphs for Multi-Scalar Urban Analytics. *In Proceedings of the 19th International Conference on Computational Urban Planning and Urban Management*, (2025, June).
- [3] Miller, C., Quintana, M., Frei, M., ... **Yap, W.**, Biljecki, F. Introducing the cool, quiet city competition: predicting smartwatch-reported heat and noise with digital twin metrics. *In Proceedings of the 10th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation*, 298-299, (2023, November).
- [2] **Yap, W.** Biljecki, F. Global Tool for Open Urban Network Analysis. *UN-Habitat Urban Thinkers Campus Workshop for FOSS4G Asia*, (2023, December).
- [1] **Yap, W.** Neo, H. Drivers and Challenges of Digital Citizen Participation: Experiential Evidence from Southeast Asian Cities. *RC21 Conference on Sensing the City*, (2021, July).

CASE STUDY REPORTS

- [1] Chan, H. C. **Yap, W.** Housing and the city: The Singapore case. *Council of Urban Initiatives, UN-Habitat, UCL IIPP and LSE Cities*, (2023).

BOOKS AND BOOK CHAPTERS

- [7] Nicholas, S. **Yap, W.** Is the City Center Hollowing Out?, In Chan, H. C., Neo, H., Joyce, S (Eds), *Socio-spatial Reconfiguration of the City Post-Covid*, *World Scientific*, (2024).
- [6] Chan, H. C., **Yap, W.**, Nicholas, S. The Magic of Modernity, In Chan, H. C., Neo, H., Joyce, S (Eds), *Socio-spatial Reconfiguration of the City Post-Covid*, *World Scientific*, (2024).
- [5] Yuen, B., Močnik, Š., Moogoor, A., Dieterich, A., **Yap, W.**, Bhuyan, M. R. Age-friendly Neighbourhood Environmental Audit Toolkit, *World Scientific*, (2022).
- [4] Yuen, B., Bhuyan, M. R., Dieterich, A., **Yap, W.**, Chua, R. Age-friendly Neighbourhood Post-implementation Toolkit, *World Scientific*, (2022).
- [3] Yuen, B., Bhuyan, M. R., Song, S., Moogoor, A., **Yap, W.**, Močnik, Š., Chua, R. Age-friendly Neighbourhood Planning and Design Guidelines: A Singapore Case Study, *World Scientific*, (2022).
- [2] **Yap, W.** Norakmal, H. Tokyo - Pushing Frontiers, Planning for Society 5.0., In Arturo, B., Christos, C., Bruno, L., Chan, H. C. (Eds), *Cities in a Time of Global Emergencies*. *IMD Business School*, (2021).
- [1] Yuen, B., Močnik, Š., Yu, C. F., **Yap, W.** Ageing-friendly neighbourhoods in Singapore, Asia-Pacific, Europe and North America: An Annotated Bibliography, *Springer Briefs in Aging*, (2020).

OPEN-SOURCE SOFTWARE

- [1] **Yap, W.** (Lead Developer). Urbanity: Open-Source Python library for Urban Graph Analytics. (164 stars). *GitHub* 2022. <https://github.com/winstonyym/urbanity>

OPEN-ACCESS DATASETS

- [2] **Yap, W.** Biljecki, F. A Global Feature-Rich Network Dataset of Cities and Dashboard for Comprehensive Urban Analyses. *Figshare* 2023. <https://doi.org/10.6084/m9.figshare.22124219>
- [1] Hou, Y., Quintana, M., Khomiakov, M., **Yap, W.**, Ouyang, J., Ito, K., ... Biljecki, F. Global Streetscapes-A comprehensive dataset of 10 million street-level images across 688 cities for urban science and analytics. *Hugging Face* 2024. <https://huggingface.co/datasets/NUS-UAL/global-streetscapes>

PRESENTATION AND SESSIONS

DEPARTMENT AND LAB SEMINARS

- [6] “Urban Graph Analytics: Connecting Cities, Data, and People”, at the *Institute of Remote Sensing and GIS*, Peking University, October 2025.
- [5] “Urban Graph Analytics: Connecting Cities, Data, and People”, at the *Department of Architecture*, Tsinghua University, October 2025.
- [4] “Leveraging Urban Graphs for Multiscalar Urban Analytics”, at the *Sustainable Design Group*, University of Cambridge, July 2025.
- [3] “Modeling Complex Urban Systems with Open Source Software”, at the *City Form Lab*, Massachusetts Institute of Technology, October 2024.
- [2] “Multi-scalar Urban Analytics with Graphs” at the *London-Singapore Urban Analytics Seminar for Liveable Cities*, August 2025.
- [1] “Urban Network Modelling and Analytics” at the *Department of Architecture and Building Engineering*, Tokyo Institute of Technology, April 2024.

INVITED TALKS

- [8] “Scaling Urban Analytics Globally with Open Data and Open-Source Tools”, at *University of Southern California, Spatial Sciences Institute*, April 2026.
- [7] “Urban Health and Graph Deep Learning in US Cities”, at *Imperial College London School of Public Health*, February 2026.
- [6] “Unleashing the Power of Connections in Cities with Graphs”, at *Cambridge Centre for Advanced Research and Education*, December 2025.
- [5] “Revealing the Power of Connections in Cities: Urban Graphs for Scalable, Data-Driven Planning and Design”, at *Lee Kuan Yew Centre for Innovative Cities Seminar Series*, September 2025.
- [4] “Open Source Software & Urban Network Analytics”, at *Takenaka Corporation Tokyo Headquarters*, April 2024.
- [3] “Unlocking Urban Insights: How Technology Shapes City Planning and Management”. *Singapore Management University Integrative Fiesta*, September, 2023.
- [2] “Multidimensional Network Modelling with Urbanity”. *127th OGC Member Meeting: Digital Twins Special Session*, September 2023.
- [1] “Free and Open Source Digital Planning Tools”. *Planning Institute of Australia, PlanTech Series*, July 2023.

CONFERENCE PRESENTATIONS

- [4] “Leveraging Urban Graphs for Multi-Scalar Urban Analytics” *19th International Conference on Computational Urban Planning and Urban Management*, June 2025.
- [3] “Science-based Approach to Future Scenario Planning” *Urban Solutions and Sustainability R&D Conference*, October 2023.
- [2] “Free and Open Source Urbanism: Software for Urban Planning” at the *NUS-SUTD PhD Symposium*, May 2022.
- [1] “Drivers and Challenges of Digital Citizen Participation: Experiential Evidence from Southeast Asian Cities” at the *RC21 Conference: Sensing the City*, July 2021.

ORGANIZED PANELS

[2] "Decarbonizing Cities" in *International Forum of Urbanism Global Seminar*, October 2024.

Panellists: Prof. Arno Schlueter (ETH Zurich) and Prof. Jin Yeu Tsou (City University of Hong Kong)

[1] "Street view imagery: Have we answered all the questions with it What's left to do" in *4th Spatial Data Science Symposium*, September 2023.

Panellists: Prof. Yuhao Kang (University of South Carolina), Prof. Mohamed R. Ibrahim (University of Leeds), and Prof. Pengyuan Liu (Nanjing University of Information Science and Technology)

TEACHING EXPERIENCES

Cornell University

Guest Lecturer

Spring 2026

Course: Data Science for Socio-Technical Systems (SYSEN 5460)

Teaching Faculty: Professor Timothy Fraser

- Introduce open source urban analytics to M.Eng system engineering students

National University of Singapore

Guest Lecturer

Spring 2023

Course: Introduction to Geographic Information System

Faculty: Professor Rudi Stouffs

- Design and deliver lecture to postgraduate students

Teaching Assistant

Fall 2023

Course Title: Artificial Intelligence for Design

Teaching Faculty: Professor Rudi Stouffs

- Led programming tutorial lecture
- Mentored undergraduate students in open-ended projects.

Teaching Assistant

Spring 2022

Course Title: Planning Technologies and Spatial Programming

Teaching Faculty: Professor Filip Biljecki

- Help code troubleshooting during lectures
- Mentored graduate students in open-ended projects
- Project evaluation and grading

Singapore University of Technology and Design

Guest Lecturer

Spring 2023

- Helped design the course, including lectures, homework, and project
- Led programming workshop and lecture

STUDENTS MENTORED

National University of Singapore

- Master Student in Spatial, Transport and Environmental Economics: Brian van Laar (cross-registered from VU Amsterdam)
- Master Student in Urban Planning: Gu Youlong

Singapore University of Technology and Design

- Bachelor of Engineering in Computer Science: Joshua Tay
- Bachelor of Social Sciences in Geography: Anniesa Foo

PROFESSIONAL SERVICES & LEADERSHIP

Organizing Committee

- College Representative, NUS Heart of Leadership Programme, 2025
- Singapore Representative, United Nations Industrial Development Organization Bridge for Cities Dialogue, 2025
- Young Leader, World Cities Summit Young Leaders Symposium, 2024 and 2025
- Singapore Representative, Japanese Education Ministerial Roundtable Dialogue, 2024
- Session Chair, International Forum of Urbanism Global Seminar, 2024
- Workshop Chair, UN-Habitat Urban Thinker Campus Workshop, 2023
- Lead Representative, NUS Cities Panel of Advisors Showcase, 2023
- Session Chair, 4th Spatial Data Science Symposium, 2023
- Executive Committee, NUS-SUTD PhD Symposium, 2022
- College Representative, National University of Singapore Graduate Education Virtual Open House, 2022

Reviewer

- *Cities* (Elsevier)
- *Computers, Environment and Urban Systems* (Elsevier)
- *Environment and Planning B: Urban Analytics and City Science* (SAGE)
- *Geographical Analysis* (Wiley)
- *Geoforum* (Elsevier)
- *International Journal of Applied Earth Observation and Geoinformation* (Elsevier)
- *International Journal of Digital Earth* (Taylor & Francis)
- *International Journal of Geographical Information Science* (Taylor & Francis)
- *Journal of Open Source Software*
- *npj Urban Sustainability* (Springer Nature)
- *Nature Climate Change* (Springer Nature)
- *Nature Cities* (Springer Nature)
- *Scientific Data* (Springer Nature)
- *Scientific Reports* (Springer Nature)
- *Sustainable Cities and Society* (Elsevier)
- *Transactions in GIS* (Wiley)
- *Transportation Research Part D: Transport and Environment* (Elsevier)
- *Urban Informatics* (Springer Nature)

OUTREACH ACTIVITIES

Singapore-Nagoya Overseas Partnership 2025

<https://www.nuglobalcampus.org/>

- Led and organize research exchange with postgraduate students and staff

NUS Public Events Series 2025

https://cde.nus.edu.sg/arch/news_and_events/events_ay2526_public_lecture_convo_260825/

- Led and organize lab visit with Professor Perry Yang (Georgia Tech) and Professor Miho Mazereeuw (MIT)

London-Singapore Urban Analytics Joint Seminar for Liveable Cities 2025

<https://ual.sg/seminars/>

- Coordinate exchange and research presentation

Cambridge-NUS Joint Seminar 2025

<https://www.sustainabledesign.arct.cam.ac.uk/news/joint-seminar-day-nus-urban-analytics-lab-27th-june-2025>

- Led and organize research exchange with postgraduate students and staff

Singapore-Chulalongkorn Collaboration 2023

<https://ual.sg/post/2023/07/28/visit-by-uddc-from-chulalongkorn-university/>

- Facilitate research exchange with postgraduate students and staff

Singapore Institute of Planners Mentorship Programme 2023
<https://www.planning.org.sg/nw/20250106/SIP-Mentor-Program2.pdf>
• Experience sharing and engagement with young planners

NUS Real Estate Ambassador Programme 2016
<https://studylib.net/doc/13142814/nus-real-estate-news>
• Lead highschool outreach activities on urban planning and the built environment
• Organize campus open house booth and activities

MEDIA COVERAGE

MIT News 2026
"How urban design leads to better wellness"
<https://news.mit.edu/2026/how-urban-design-leads-to-better-wellness-0701>

College of Design and Engineering Research 2025
"AI model maps building emissions to drive smarter, fairer climate solutions | CDE Research"
https://www.youtube.com/watch?v=rushehhq_iI

Urban Analytics Lab 2025
"Building smarter, more sustainable, liveable cities: The Urban Analytics Lab"
<https://vimeo.com/764033095>

ASHRAE Insights Newsletter 2025
"AI model maps building emissions to support fairer climate policies"